

Lab Report: Week 8 Session B

Final Demo & Defense Preparation

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May 22, 2026

Abstract

This report documents Week 8 Session B: **final demo and defense preparation** for the Smart Water capstone. Deliverables include a 5-minute `PRESENTATION_OUTLINE.md`, `DEMO_SCRIPT.md` for live Streamlit rehearsal, `JAGGED_FRONTIER.md` reflection, and `DELIVERABLES_CHECKLIST.md`. GitHub commit `1ada4e6` (*Week 8 Session B: presentation and jagged frontier deliverables*) is on `main`. Evidence: Figures 1a–1b. The integrated app runs via `streamlit run app/main.py`; tests: 29 passed, 96% coverage (Session 8A).

1 Introduction

Week 8 Session B follows implementation (Week 7B) and validation (Week 8A). The focus shifts from writing code to **communicating** the capstone: a timed presentation, rehearsed demo, organized submission package, and reflection on AI strengths and limits (Jagged Frontier).

2 Objectives

- Prepare a 5-minute presentation structure (Exercise 1).
- Practice live demo flow (Exercise 2).
- Organize all capstone deliverables (Exercise 3).
- Final review before defense (Exercise 4).

3 Exercise 1: Presentation preparation

3.1 Five-minute outline

Table 1: Presentation timing (`PRESENTATION_OUTLINE.md`).

Block	Content	Minutes
1	Problem and integrated dashboard goal	1
2	Live Streamlit demo (four tabs)	2
3	AI collaboration (<code>prompt_log</code> , <code>AGENTS.md</code>)	1
4	Tool evaluation and Jagged Frontier	1

Problem statement (summary): Separate Week 3–6 lab scripts for rainfall, SCS-CN runoff, reservoir dispatch, and DEM flood mapping are unified in one Streamlit teaching dashboard with documented AI-assisted development.

3.2 Demo script

DEMO_SCRIPT.md defines tab order: (1) Weather/Alerts with RED at 22 mm/h peak; (2) Runoff with $Q \leq P$ check; (3) Reservoir schedule $\approx \$708,849$ PASS; (4) Flood map with monotonic flooded % vs. stage. Backup screenshots from Weeks 7–8 if live demo fails.

4 Exercise 2: Practice demo

Command:

```
cd ai_water_lab/capstone
streamlit run app/main.py
```

URL: <http://localhost:8501>

Rehearsal checklist: start app before class; test screen sharing; keep `week7_session_b.Streamlit_page.png` and `week8_session_a_terminal.png` as backups.

5 Exercise 3: Deliverables organization

Table 2: Capstone deliverables checklist (from DELIVERABLES_CHECKLIST.md).

Item	Status
GitHub repository	On <code>main</code> , 6+ commits
AGENTS.md	Present
prompt_log.md	Sessions 7A, 7B, 8A, 8B
JAGGED_FRONTIER.md	Present
README.md	Setup and run commands
Working Streamlit app	Four tabs implemented
pytest	29 passed, 96% <code>src/</code> coverage
Presentation files	Outline + demo script

5.1 GitHub evidence

Figures 1a and 1b show commit `1ada4e6` adding Session B files and the README with setup instructions.

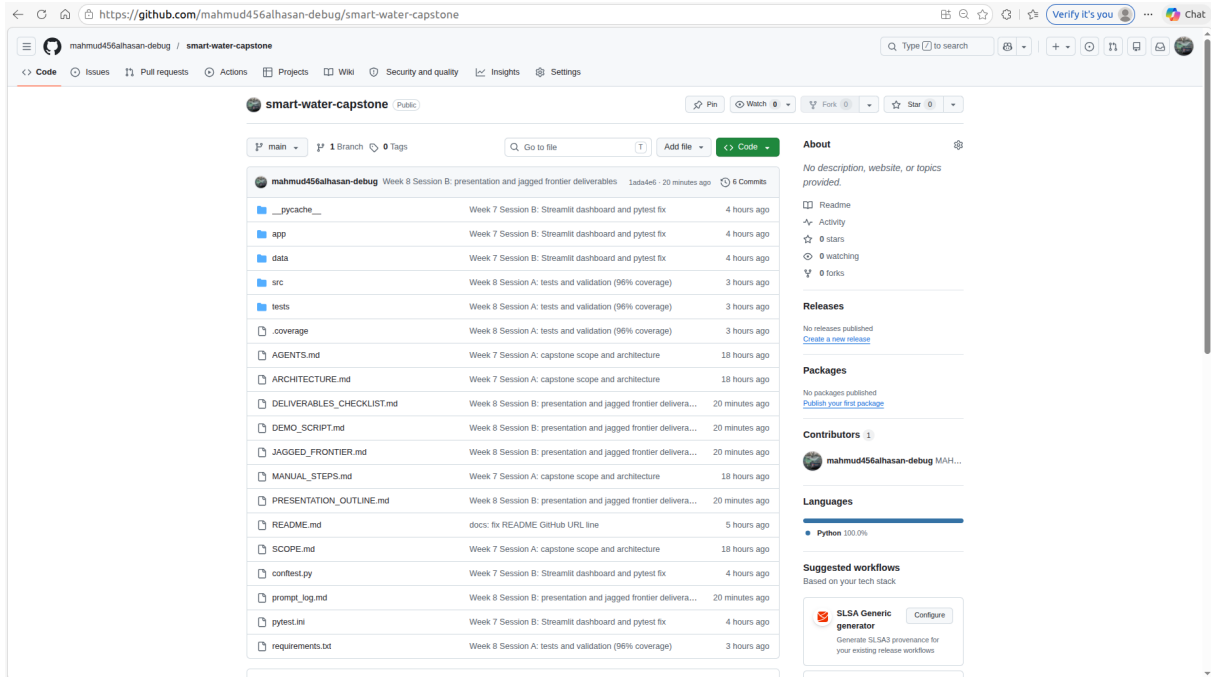
6 Exercise 4: Final review

Defense readiness: Presentation outline and demo script complete; physical validation and tests from Session 8A reduce demo risk; Jagged Frontier documents where human review remains essential.

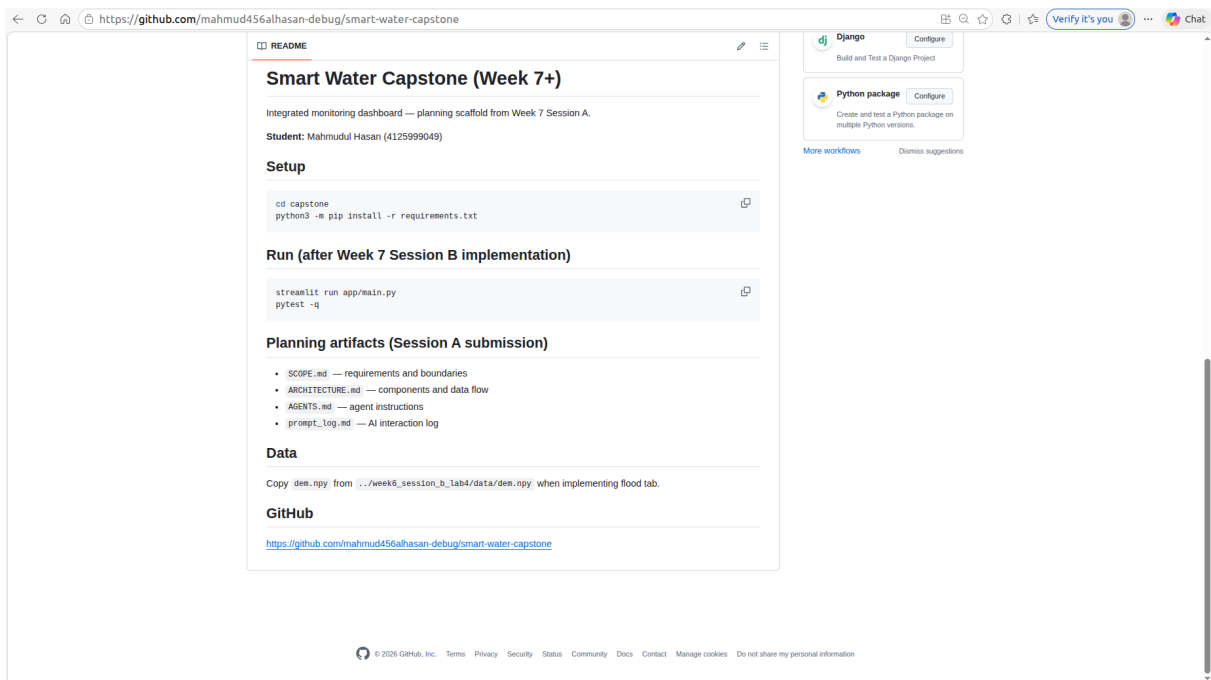
AI tools: Primary development via **Cursor Agent**; OpenCode not required. Optional cross-check with ChatGPT/Gemini for architecture (Week 7A) per course guidance.

7 Jagged Frontier summary

AI excelled at scaffolding, pytest boilerplate, and Streamlit layout. AI struggled with SCS-CN edge cases (false assumption $P=5$ mm always below I_a), mixed storage/flow units without explicit prompts, and git authentication prompts. Swiss Cheese layers (review, 29 tests, `validation.py`, integration tests) compensated. Full text in Appendix A.



(a) Repository tree: `DELIVERABLES_CHECKLIST.md`, `DEMO_SCRIPT.md`, `JAGGED_FRONTIER.md`, `PRESENTATION_OUTLINE.md`; commit history Weeks 7–8.



(b) Rendered README.md: install, `streamlit run app/main.py`, `pytest -q`, planning artifacts.

Figure 1: GitHub evidence for Exercise 3 (deliverables on <https://github.com/mahmud456alhasan-debug/smart-water-capstone>).

8 Discussion

1. Session B materials turn a code repo into a defensible capstone story.
2. Two GitHub screenshots document deliverables without requiring a separate Streamlit capture for this session.
3. Live defense remains the primary assessment; written prep reduces timing overrun.

9 Conclusion

Week 8 Session B objectives were met: 5-minute presentation plan, demo script, deliverables checklist, Jagged Frontier reflection, and GitHub push (1ada4e6). The capstone is ready for live demonstration and course submission via <https://github.com/mahmud456alhasan-debug/smart-water-capstone>.

A Jagged Frontier reflection

JAGGED_FRONTIER.md (full file)

```
# Jagged Frontier Reflection -- Smart Water Capstone

**Student:** Mahmudul Hasan (4125999049)

The **Jagged Frontier** means AI is strong on some tasks and weak on others -- not
↳ uniformly "smart" or "dumb."

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## Where AI excelled (high frontier)

| Task | Example in this project |
|-----|-----|
| Scaffolding | Week 7A: `SCOPE.md`, folder tree, `AGENTS.md`, README in one session |
| Repetitive code | Streamlit tab layout, pytest templates, CSV loader |
| Known patterns | SciPy `minimize_scalar` reservoir loop from lab guide |
| Documentation | Architecture diagrams, prompt log structure |

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## Where AI struggled (low frontier)

| Task | Example | How we caught it |
|-----|-----|-----|
| Physical edge cases | Assumed P=5 mm always gives zero runoff for CN=95 | **pytest  
↳ failed**; Ia is smaller for high CN |
| Unit consistency | MCM storage vs m3/s flows (Week 6) | Manual review + lab guide;  
↳ `feasible_release_bounds` fix |
| Test coverage gaps | 27% with only 2 tests | Coverage report showed 0% on  
↳ weather/reservoir |
| Git workflow | Wrong username at `git push` prompt | Human read terminal errors |

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## Swiss Cheese defense (how we compensated)

1. **Human review** -- units and hydrology rules before trusting merges
2. **Unit tests** -- 29 tests, 96% coverage on `src/`
3. **Physical validation** -- `src/validation.py` (Q<=P, storage bounds, monotonic flood)
4. **Integration tests** -- rainfall -> alert, full reservoir wrapper
```

```

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## Tool comparison (short)

| Tool | Best for this capstone |
|-----|-----|
| Cursor Agent | Implementation, tests, reports in-repo |
| ChatGPT / Gemini | Optional second opinion on architecture (Week 7A) |
| OpenCode | Not required; useful in Week 1-2 for AGENTS.md pattern |

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## What I would do differently

- Add `pytest.ini` and validation layer before calling the app "done."
- Document unit conventions in the first prompt, not after bugs.
- Rehearse demo script once with a timer before defense.

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## Conclusion

AI accelerated the capstone but did not replace domain checks. The jagged frontier showed
↪ up in small hydrology assumptions, not in writing another Streamlit button. Tests
↪ and validation were the safety net.

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B Prompt log (Session 8B excerpt)

prompt_log.md (Session 8B section)

```

# Week 8 Session B -- Final Demo & Defense Prep

Tool: Cursor Agent | Date: 2026-05-22

## Exercise 1: Presentation

Created: `PRESENTATION_OUTLINE.md` (5 min: intro, demo, AI, lessons),
↪ `DEMO_SCRIPT.md` (four Streamlit tabs).

## Exercise 2: Practice demo

Command: `streamlit run app/main.py` at http://localhost:8501

Dry run: (fill in after you time yourself: ___ minutes)

## Exercise 3: Deliverables

Created: `JAGGED_FRONTIER.md`, `DELIVERABLES_CHECKLIST.md`

Repo: https://github.com/mahmud456alhasan-debug/smart-water-capstone

## Exercise 4: Final review

Ready: AGENTS.md, README, prompt_log, tests, app, presentation files.

Lessons learned: Rehearse with timer; keep pytest screenshot and Streamlit screenshot
↪ as backup.

```